

BELIMO

Message Decipher Tool

(Belimo_Message_Decipher_Tool.xlsm)

USER MANUAL

Release 1.5

Prepared for Belimo internal staff – not for distribution.

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Questions? Use the Report a Bug or Give Feedback button in the tool, or contact the tool's maintainer.

Contents

1. Introduction	4
1.1 What the tool does, in one paragraph	4
1.2 What's in this manual	4
1.3 Supported products and versions	5
2. Getting Started	6
2.1 Opening the file	6
2.2 The About screen	6
2.3 How updates work	7
3. How Message Codes Work	8
3.1 One number, many conditions	8
3.2 A worked example by hand	8
3.3 Bit values	8
3.4 Where to read the code from	8
4. The Calculator Sheet	10
4.1 The controls	11
4.2 Decoding a code, step by step	11
4.3 Worked example: Energy Valve EV4, code 164	11
4.4 Reading the result table	12
4.5 Code types	13
4.6 More examples	13
5. BACnet Pic PDFs and Feedback	15
5.1 BACnet Pic PDF	15
5.2 Report a Bug or Give Feedback	15
6. The Reference Sheets	16
6.1 The code tables (EVCodes, ePIVCodes, TEMCodes)	16
6.2 EV Compare	16
6.3 The BACnet interface sheets	17

7. Messages the Tool Can Show You	18
8. Troubleshooting and Getting Help	19
Appendix A. Message Tables	20
A.1 Energy Valve	20
A.2 ePIV	25
A.3 TEM	28
Appendix B. Release History	29

1. Introduction

The Belimo Message Decipher Tool (the “tool”, file name Belimo_Message_Decipher_Tool.xlsm) is an Excel workbook that turns a single status number read from a Belimo device into a plain-English list of what is wrong. Belimo Energy Valves, ePIV pressure-independent valves and TEM thermal energy meters all report their alarms, warnings and status flags in a compact form: one whole number in which each individual condition is a single “bit”. That number is easy for a controller to store and almost impossible for a person to read. You type the number in, tell the tool which product and software version it came from, and it lists every condition that is switched on, with a diagnostic explanation for each.

Who this manual covers

Belimo sales, application and service staff who read a status or error code from a BACnet object, a Webview page or an Energy Valve data file (.csv) and need to know what it means. No knowledge of binary numbers is needed – Section 3 explains the idea in plain language, and the tool does the arithmetic for you.

1.1 What the tool does, in one paragraph

You choose a **Product** (Energy Valve, ePIV or TEM), the **Version** of its software, and the **Code Type** you are decoding (an Error State, or a Non-Persistent or Persistent error where the device has them). You enter the **Message Code** – the whole number you read from the device – and click Calculate. The tool splits the number into its individual bits and shows, for each bit that is set, the bit number (Message ID), its numeric value, the message the device is reporting, and a diagnostic note describing the likely cause. A running Bit Total confirms that the listed bits add back up to the number you entered.

1.2 What’s in this manual

- Section 2 – Getting started: opening the file, enabling macros, and the About screen (updates and this manual).
- Section 3 – How message codes work, and where to read the code from on each product.
- Section 4 – The Calculator sheet: every control, step by step, with worked examples.
- Section 5 – The BACnet Pic PDFs and the Feedback button.
- Section 6 – The reference sheets inside the workbook (code tables, EV Compare, and the BACnet interface sheets).
- Section 7 – Messages the tool can show you, and how to fix them.
- Section 8 – Troubleshooting and getting help.
- Appendix A – The complete message tables for every supported product, version and code type.
- Appendix B – Release history.

1.3 Supported products and versions

Product	Version (dropdown)	Object that carries the code	Highest valid code
Energy Valve	EV2	AI[100]	511
Energy Valve	EV3	AI[100]	4095
Energy Valve	EV4	AV[140] (Error State)	16383
Energy Valve	EV4.2	AV[140] (Error State)	524287
ePIV	ePIV2	MI[106]	15
ePIV	ePIV3	MI[110]	15
ePIV	ePIV4	AV[140] (Error State)	65535
TEM	TEM4	AV[140]	8191
TEM	V4.2_MV[103]	MV[103] Status Sensor	63
TEM	V4.2_MV[104]	MV[104] Status Flow	63
TEM	V4.2_MV[105]	MV[105] Status Media	63

EV4, EV4.2 and ePIV4 additionally have **Non-Persistent** and **Persistent** error codes, which are selected with the Code Type dropdown (Section 4.5). The “highest valid code” is the largest number the tool will accept for that version; anything larger is rejected (Section 7).

2. Getting Started

2.1 Opening the file

Open Belimo_Message_Decipher_Tool.xlsm in Microsoft Excel. The tool is built with macros, so Excel must be allowed to run them: if Excel shows a yellow **Protected View** bar, click **Enable Editing**, and if it then shows a **Security Warning** bar, click **Enable Content**. (If the file came from e-mail or a download, right-click it in File Explorer, choose Properties, tick **Unblock** and click OK before opening it – this avoids the warnings entirely.)

Each time the workbook opens it does two things automatically. First it quietly checks Belimo's update source to see whether a newer release exists (Section 2.3). Second it resets the Calculator sheet: the Product, Version and Code Type boxes go back to “-Select-” and any earlier result is cleared, so you always start from a clean sheet.

2.2 The About screen

Click the small round **(i)** button in the top-right corner of the Calculator sheet (next to the picture of the magnifying glass) to open the About screen. It shows the tool's name, its version number and release date, the internal-use notice, a short description, and who maintains the tool. It also has three buttons and a Close button.



Figure 1 – The About screen.

Button	What it does
Check for Updates	Asks the update source whether a newer release exists. If one does, you are told what is new and may choose to download it.
Download Fresh Copy	Downloads the current published release straight away, even if you are already up to date – useful if you think your copy is damaged or missing something.
Download User Manual	Saves the current published copy of this manual (a PDF) to a folder you choose. If no manual has been published yet, the tool tells you so.

Button	What it does
Close	Closes the About screen.

2.3 How updates work

When a newer release is available the tool tells you and offers to show what has changed. If you choose to install it, you pick where the new copy is saved through a normal Windows **Save As** dialog – exactly like saving any other file. Your current copy is never changed or deleted, so there is nothing to lose by trying it. If your computer cannot reach the update source (for example you are offline), the check simply does nothing and the tool works as normal.

3. How Message Codes Work

3.1 One number, many conditions

A Belimo device can have several conditions active at once – for example the actuator cannot move *and* there is flow with the valve closed. Instead of a separate object for each, the device keeps one whole number and gives every possible condition its own **bit**: bit 0 is worth 1, bit 1 is worth 2, bit 2 is worth 4, bit 3 is worth 8, and each further bit is worth double the one before (16, 32, 64, 128, 256 ...). When a condition is active, its value is added into the number.

Reading the number backwards is therefore a matter of working out which of those values were added. Every whole number can be made in exactly one way, so there is never any ambiguity.

3.2 A worked example by hand

Suppose an EV4 Energy Valve reports the code **164**. Starting with the largest value that fits:

Step	Remaining	Largest bit value that fits	Bit	Left over
1	164	128	7	36
2	36	32	5	4
3	4	4	2	0

So $164 = 128 + 32 + 4$, which means bits 7, 5 and 2 are set. Looking those bits up in the EV4 table gives *Flow measurement error*, *Flow with closed valve* and *Actuator cannot move*. This is exactly what the tool does when you click Calculate (Figure 3 in Section 4.3 shows the result).

3.3 Bit values

Bit	0	1	2	3	4	5	6	7	8	9
Value	1	2	4	8	16	32	64	128	256	512

Bit	10	11	12	13	14	15	16	17	18
Value	1024	2048	4096	8192	16384	32768	65536	131072	262144

3.4 Where to read the code from

The tool only decodes a number; you have to get the number from the device first. Where it lives depends on the product and software version:

- **Energy Valve EV2 and EV3** – BACnet analog input AI[100] (the error/status word).
- **Energy Valve EV4 and EV4.2** – BACnet analog value AV[140] for the Error State code. The Non-Persistent and Persistent error words are decoded with the same tool by choosing the matching Code Type.

- **ePIV 2** – BACnet multistate input MI[106] (StatusActuator). **ePIV 3** – MI[110]. **ePIV 4** – AV[140].
- **TEM 4** – AV[140]. **TEM 4.2** – three separate status objects: MV[103] (Status Sensor), MV[104] (Status Flow) and MV[105] (Status Media); decode each one on its own by choosing the matching version.
- **Energy Valve data files (.csv)** – the message code values recorded in a downloaded data file are decoded the same way.

Not sure which software version you have?

Click the small (i) button next to the Version box. It reminds you to check the **Application Software Version** in Webview or in the Belimo Assistant app. Choosing the wrong version decodes the right number against the wrong table, so the answer will look plausible but be wrong.

4. The Calculator Sheet

The Calculator sheet is the only sheet you need for day-to-day decoding. Figure 2 shows it as it looks when the workbook first opens.

The screenshot shows the 'Belimo Message Decipher Tool' interface, version 1.4. It features a dark blue header with the title and version. Below the header, there are five steps for decoding a message, each with a corresponding button: Step 1: Select a Product, Step 2: Select a Product Version, Step 3: Select the Message Type, Step 4: Enter the Message Code, and Step 5: Click the Calculate Button. To the left of these steps are dropdown menus for Product, Version, and Code Type, and a text input field for the Message Code. A 'Calculate' button is located below the Message Code field. To the right of the steps is a magnifying glass icon over a background of binary code. Below the steps, there is a table with four columns: Message ID, Bit Value, Message, and Diagnostics. The table has 10 rows, with the first row highlighted in orange. At the bottom left, there is a 'Bit Total' field showing 0. At the bottom right, there are two buttons: 'BACnet Pic PDF' and 'Feedback'.

Message ID	Bit Value	Message	Diagnostics

Bit Total: 0

BACnet Pic PDF Feedback

Figure 2 – The Calculator sheet on opening, with nothing selected.

4.1 The controls

Control	What it is for
Product	Energy Valve, ePIV or TEM. Choosing a product fills in the Version list and clears any earlier result.
Version	The software version of the device (see Section 1.3). Choosing a version fills in the Code Type list and clears any earlier result.
Code Type	Error State, Non-Persistent or Persistent. Only the types that exist for the chosen version are offered (Section 4.5).
Message Code (orange box)	Type the whole number read from the device. Digits only – no commas, decimal points or spaces.
Calculate button	Decodes the code and fills in the result table.
Step 1–5 labels	Reminders of the order to work in. They are labels, not buttons.
(i) next to Version	Shows a reminder of where to find the software version (Section 3.4).
(i) top right	Opens the About screen (Section 2.2).
Result table	Message ID, Bit Value, Message and Diagnostics – one row for every bit that is set.
Bit Total	The sum of the Bit Values in the table. It should equal the code you entered.
BACnet Pic PDF, Download	Opens Belimo's BACnet interface document for a product version (Section 5.1).
Report a Bug or Give Feedback	Opens Belimo's bug and feedback form in your web browser (Section 5.2).

4.2 Decoding a code, step by step

1. Click the **Product** dropdown and choose Energy Valve, ePIV or TEM.
2. Click the **Version** dropdown and choose the software version of the device.
3. Click the **Code Type** dropdown. For most versions there is only one choice, Error State; choose it.
4. Click the orange **Message Code** box and type the number you read from the device.
5. Click **Calculate**.
6. Read the table: each row is one active condition. Check that the **Bit Total** equals the number you typed.

You can decode another code for the same product simply by typing a new number and clicking Calculate again – the earlier result is cleared first. Changing the Product or Version also clears the table.

4.3 Worked example: Energy Valve EV4, code 164

Product *Energy Valve*, Version *EV4*, Code Type *Error State*, Message Code *164* gives three rows:

Figure 3 – Result for an EV4 Error State code of 164. Bit Total 164 confirms the three bits add back to the entered code.

Reading the rows top to bottom (largest bit first): **Message ID 7** (value 128) is a *Flow measurement error* – air in the system, an error occurred during flow measurement; **Message ID 5** (value 32) is *Flow with closed valve* – flow is measured although the valve is closed; and **Message ID 2** (value 4) is *Actuator cannot move* – mechanical overload due to a blocked valve. The status line under the table reads “Calculating Complete!” when the tool has finished.

4.4 Reading the result table

Column	Meaning
Message ID	The bit number (0, 1, 2 ...). This is the number used in Belimo's documentation for that status word.
Bit Value	What that bit is worth in the code (1, 2, 4, 8 ... = 2 to the power of the Message ID).
Message	The short name of the condition, as published for that product and software version.
Diagnostics	A fuller explanation of what the condition means or what causes it. Some bits have no separate diagnostic text and the cell is blank.

Rows are listed from the highest bit to the lowest. Up to 19 rows can appear (the most bits any version has), and the list is shorter when fewer conditions are active. If a bit is set that the tool's table has no name for, its Message cell is left blank – the Bit Total still counts it, so you can tell something is set.

4.5 Code types

Product / Version	Code types offered	What they are
EV2, EV3, ePIV2, ePIV3, TEM4, TEM 4.2	Error State	The single status word for the device.
EV4, EV4.2, ePIV4	Error State	The live error state word (AV[140]): conditions that are active now.
EV4, EV4.2, ePIV4	Non-Persistent	Internal measurement and processing faults that clear themselves when the cause goes away (for example a transit-time measurement timeout).
EV4, EV4.2, ePIV4	Persistent	Faults that are stored and stay set until cleared (for example an integrity check failure or a sensor resistance at its limit for several successive measurements).

The complete text for every code type is in Appendix A.

4.6 More examples

ePIV 4, Non-Persistent, code 276. $276 = 256 + 16 + 4$, so bits 8, 4 and 2 are set. Bit 8 is *calculation error* and bit 2 is *transit time out of range*. Bit 4 has no published Non-Persistent message, so its row is shown with the bit number and value but no text; the Bit Total of 276 shows nothing was missed.

[illegible]

Figure 4 – ePIV 4, Non-Persistent, code 276.

TEM 4, code 2120. 2120 = 2048 + 64 + 8: *Freeze Warning* (bit 11), *Flow Actual exceeds Flow Nominal* (bit 6) and *Reverse Flow* (bit 3).

Energy Valve EV4.2, code 262147. $262147 = 262144 + 2 + 1$: *Minimum position applied* (bit 18), *Gear Disengaged* (bit 1) and *No Communication with Actuator* (bit 0).

Energy Valve EV3, code 2049. $2049 = 2048 + 1$: *Freeze Warning* (bit 11) and *Error Sensor T1* (bit 0). EV3 codes run up to 4095, so bit 11 is the highest.

TEM 4.2, MV[103] Status Sensor, code 6. $6 = 4 + 2$: *Flow Measurement Error* (bit 2) and *OK* (bit 1).

5. BACnet Pic PDFs and Feedback

5.1 BACnet Pic PDF

Below the result table is a **BACnet Pic PDF** section. Choose a document from its dropdown list and click **Download**: your web browser opens Belimo's BACnet interface document ("BACnet Pic") for that product version. The list changes with the Product you have selected:



Bit Total: 3
Calculating Complete!

BACnet Pic PDF Feedback

Select PDF

Download

Report a Bug or Give Feedback

Figure 5 – The BACnet Pic PDF and Feedback section below the result table.

Product	Documents offered
Energy Valve	EV_V3, EV_V3.2, EV_V4.1, EV_V4.2
ePIV	ePIV_V3.03, ePIV_V3.04, ePIV_V4.0, ePIV_V4.1, ePIV_V4.2
TEM	TEM_V4.1, TEM_V4.1.1, TEM_V4.2 (listed, but no download link is set up yet – the tool tells you so)

The documents are stored on Belimo's internal SharePoint, so you must be signed in to your Belimo account to open them. If you choose "Select PDF" (or an item with no link) and click Download, the tool shows "No website associated with this item!".

5.2 Report a Bug or Give Feedback

Click the orange **Report a Bug or Give Feedback** picture to open Belimo's bug and feedback form in your web browser. Use it for anything from a typo to a request for a new product version or code table. If you are reporting a wrong or missing message, please include the product, version, code type and the code you entered – with those the problem can usually be reproduced in seconds.

6. The Reference Sheets

Behind the Calculator, the workbook keeps the tables it decodes against, plus Belimo's BACnet interface descriptions. They are there to look things up; you do not need to open them to use the tool, and you should not edit them, because the Calculator reads from them directly. Switch between sheets with the tabs along the bottom of the Excel window.

Sheet	What it contains
Calculator	The decoding screen described in Section 4.
EVCodes	The message and diagnostic text for every bit of every Energy Valve version and code type, plus the small version table at the bottom (Section 6.1).
EV Compare	The BACnet objects of EV 2.0, 3.0, 3.2, 4.0 and 4.2 side by side (Section 6.2).
ePIVCodes	The same kind of table for ePIV 2, 3 and 4.
TEMCodes	The same kind of table for TEM 4 and TEM 4.2 (MV[103], MV[104], MV[105]).
ePIV 02.00, ePIV 03.03, ePIV 04.2.0	Belimo's BACnet object description for those ePIV software versions.
TEM 04.00	Belimo's BACnet object description for TEM version 4.00.
(Config)	A hidden sheet holding the dropdown lists, the BACnet Pic document list, and the release number and date. It is hidden on purpose.

6.1 The code tables (EVCodes, ePIVCodes, TEMCodes)

Each code sheet has one column pair per version: the bit's message, then its diagnostic text. Down the left is the bit number and the value it is worth. Below the table is a small block with the **Version**, its highest bit (**Bit CT**) and its highest valid code (**EXP**); the Calculator uses this block to fill the Version dropdown and to decide how far to work down.

Bit	^2	EV 2.0		EV 3.0		EV 4.0		EV 4.0	
		AI[100]	Error Description	AI[100]	Error Description	AV[140]	Error Description	AV[140]	Error Description
Bit 0	1	Error Sensor T1	Temp is <-15°F or > 300°F Temp is <-26°C or > 149°C	Error Sensor T1	No connection to remote temperature Sensor	No Communication with Actuator	Communication with Actuator not Possible		
Bit 1	2	Error Sensor T2	Temp is <-15°F or > 300°F Temp is <-26°C or > 149°C	Error Sensor T2	No connection to remote temperature Sensor	Gear Disengaged	Gear button is pressed		
Bit 2	4	Error Flow Sensor	For 1/2" through 2" - Check Error byte	Error Flow Sensor	Mechanical Overload due to blocked Valve	Actuator cannot move	Mechanical Overload due to blocked Valve		
Bit 3	8	Actuator cannot move	Mechanical Overload due to blocked Valve	Actuator cannot move	Mechanical Overload due to blocked Valve	Reverse Flow	Reverse flow detected		
Bit 4	16	Flow with closed valve	Flow is measured but valve is closed	Flow with closed valve	Flow is measured but valve is closed	Flow Setpoint not reached	Setpoint cannot be reached within 15 min during flow control		
Bit 5	32	Air bubbles	Air in System, error occurred during flow measurement	Air bubbles	Air in System, error occurred during flow measurement	Flow with closed valve	Flow is measured but valve is closed		
Bit 6	64	Flow not reached	Setpoint cannot be reached within 15 min during flow control	Flow not reached	Setpoint cannot be reached within 15 min during flow control	Flow actual exceeds flow nominal	Actual flow exceeds the design nominal flow		
Bit 7	128	Power not realized	Setpoint cannot be reached within 15 min during Power Control	Power not realized	Setpoint cannot be reached within 15 min during Power Control	Flow measurement error	Air in System, error occurred during flow measurement		
Bit 8	256	Gear disengaged	Gear button is pressed	Gear disengaged	Gear button is pressed	Remote Temperature not OK	No connection to remote temperature Sensor		
Bit 9	512			Reverse Flow Detected	Reverse flow detected	Flow body temperature not OK	Error with embedded sensor		
Bit 10	1024			MP Communication Faulty	interrupted	Communication to sensor interrupted	Internal communication to the flow sensor interrupted		
Bit 11	2048	4095		Freeze Warning	Measured Temperature and Glycol concentration indicate ice can build up	Freeze Warning	Measured Temperature and Glycol concentration indicate ice can build up		

Figure 6 – EVCodes sheet: bit numbers, values, and the messages for EV 2.0, 3.0 and 4.0.

6.2 EV Compare

EV Compare lists the BACnet objects – object name, type and instance, description, values, default and access – for each Energy Valve generation next to each other, so you can see at a glance what changed between EV 2.0, 3.0, 3.2, 4.0 and 4.2. Scroll right to move from one generation to the next.

EV 2.0							
Object Name	Object Type / Instance	Description	Values	Default	Access	Object Name	
Device_Name	Device [x]					Device_Name	
Analog Inputs							
RelPos	AI [1]	Relative Position in %	0...100	-	R	RelPos	
AbsPos	AI [2]	Absolute Position in °	0...90		R	AbsPos	
						SpAnalog_V	
RelFlow	AI [10]	Relative Flow in %	0...100		R	RelFlow	
AbsFlow_SI1	AI [11]	Absolute Flow in l/min	0...100,000		R	AbsFlow_lmin	
AbsFlow_SI2	AI [12]	Absolute Flow in m3/h	0...600		R	AbsFlow_m3h	
AbsFlow_US	AI [13]	Absolute Flow in gpm	0...100,000		R	AbsFlow_gpm	
AbsFlow_SI3	AI [14]	Absolute Flow in l/s	0...100,000		R	AbsFlow_ls	
						AbsFlow_lh	
T1_SI	AI [20]	Temperature 1 (remote) in °C	-10...+120		R	T1_C	
T2_SI	AI [21]	Temperature 2 (embedded) in °C	-10...+120		R	T2_C	
DeltaT_SI	AI [22]	Delta Temperature in °C	-500...+500		R	DeltaT_K	
T1_US	AI [25]	Temperature 1 (remote) in °F	14...248		R	T1_F	
T2_US	AI [26]	Temperature 2 (embedded) in °F	14...248		R	T2_F	
DeltaT_US	AI [27]	Delta Temperature in °F	-500...+500		R	DeltaT_F	
AbsPower_SI	AI [30]	Power in kW	0...2.147e+9	-	R	AbsPower_kW	

Figure 7 – EV Compare, showing the EV 2.0 object list (the EV 3.0 list begins at the right edge).

6.3 The BACnet interface sheets

The ePIV and TEM interface sheets reproduce Belimo's BACnet object description for that version: object names, types and instances, descriptions, value ranges and defaults, and which objects can be written. Use them to find the object that carries a status code (Section 3.4).

7. Messages the Tool Can Show You

Message	Meaning and what to do
Invalid entry: Please Select a Product from the List Box	Product is still “-Select-”. Choose Energy Valve, ePIV or TEM.
Invalid entry: Please Select a Version from the List Box	Version is still “-Select-”. Choose the device’s software version.
Invalid entry: Please Select a Code Type from the List Box	Code Type is still “-Select-”. Choose Error State (or Non-Persistent / Persistent where offered).
Invalid entry in Message Code Value: Please enter a whole number with no commas or decimals.	The Message Code box is empty, contains letters, or contains a decimal point or comma. Type digits only, for example 1092 not 1,092.
The value in Message Code is not an exponent of any of the numbers.	The number is zero, negative, or far larger than any version supports (above 524287). Check that you typed it correctly. (For some very large numbers you will see the Version Limits message below instead.)
Invalid entry: The Message Code exceeds the Version Limits	The number is larger than the highest valid code for the chosen version (see the table in Section 1.3). Either the code was mistyped, or the Version is set too low – for example an EV4.2 code decoded as EV2.
Bit Value Not Found (shown in the cell beside Version)	The chosen Version was not found in the version table. Re-select the Product and Version.
No website associated with this item!	The BACnet Pic dropdown is on “Select PDF” or on an item that has no link yet.
Version Information	The (i) next to Version: check the Application Software Version in Webview or the Belimo Assistant app.

8. Troubleshooting and Getting Help

Symptom	Likely cause and fix
The dropdowns and buttons do nothing.	Macros are disabled. Click Enable Editing / Enable Content in the yellow or red bar at the top (Section 2.1).
The Calculate button shows an error, or nothing happens.	Make sure Product, Version and Code Type are all set and the Message Code is a whole number. Reopen the workbook if Excel has been left in an odd state.
The answer looks plausible but does not match the device.	The wrong Version (or Code Type) is selected. Check the Application Software Version in Webview or the Belimo Assistant app and try again.
A row has a bit number but no message.	That bit has no published text for this version. It is still part of the code; the Bit Total includes it.
The Code Type box shows “-Select-” or is empty.	Choose the Product, then the Version, then pick a Code Type from the list. Changing the Product or Version resets the Code Type list.
The BACnet Pic PDF will not open.	You need to be signed in to Belimo’s SharePoint. Sign in with your Belimo account and try again.
Check for Updates or Download reports a problem.	The update source could not be reached (offline, VPN, or a firewall). The tool keeps working; try again later.
Excel says the file is locked or read-only.	Someone (or another Excel window of yours) already has it open. Close the other copy.

If the tool ever shows an unexpected error, note what you were doing and what the message said, then use the Report a Bug or Give Feedback button on the Calculator sheet (Section 5.2). For questions this manual does not answer, contact the tool’s maintainer shown on the About screen.

Appendix A. Message Tables

These are the tables the tool decodes against, listed by product. “Bit” is the Message ID shown in the result table and “Value” is the Bit Value. A bit that does not appear in a table is unused for that version. Where the diagnostic column is blank, the tool shows no separate diagnostic text for that bit.

A.1 Energy Valve

EV2 – AI[100] (highest code 511)

Bit	Value	Message	Diagnostics
0	1	Error Sensor T1	Temp is <-15°F or > 300°F Temp is <-26°C or > 149°C Temporary Consequences: Min. dT functionality is temporary not active. Power Control is temporary not active --> Flow Control Power calculation is 0.
1	2	Error Sensor T2	Temp is <-15°F or > 300°F Temp is <-26°C or > 149°C Temporary Consequences: Min. dT functionality is temporary not active. Power Control is temporary not active --> Flow Control Power calculation is 0.
2	4	Error Flow Sensor	For ½”through 2” - Check Error byte communicates failure status. 0=OK 1=Failed. For 2½” through 6” - Flow sensor reading is below 3mA or higher than 20.5mA.
3	8	Actuator cannot move	Mechanical Overload due to blocked Valve
4	16	Flow with closed valve	Flow is measured but valve is closed
5	32	Air bubbles	Air in System, error occurred during flow measurement
6	64	Flow not reached	Setpoint cannot be reached within 15 min during flow control
7	128	Power not realized	Setpoint cannot be reached within 15 min during Power Control
8	256	Gear disengaged	Gear button is pressed

EV3 – AI[100] (highest code 4095)

Bit	Value	Message	Diagnostics
0	1	Error Sensor T1	No connection to remote temperature Sensor
1	2	Error Sensor T2	No connection to remote temperature Sensor
2	4	Error Flow Sensor	
3	8	Actuator cannot move	Mechanical Overload due to blocked Valve
4	16	Flow with closed valve	Flow is measured but valve is closed
5	32	Air bubbles	Air in System, error occurred during flow measurement
6	64	Flow not reached	Setpoint cannot be reached within 15 min during flow control
7	128	Power not realized	Setpoint cannot be reached within 15 min during Power Control
8	256	Gear disengaged	Gear button is pressed
9	512	Reverse Flow Detected	Reverse flow detected
10	1024	MP Communication Faulty	Internal communication to the flow sensor interrupted
11	2048	Freeze Warning	Measured Temperature and Glycol concentration indicate ice can builds up

EV4 – Error State, AV[140] (highest code 16383)

Bit	Value	Message	Diagnostics
0	1	No Communication with Actuator	Communication with Actuator not Possible
1	2	Gear Disengaged	Gear button is pressed
2	4	Actuator cannot move	Mechanical Overload due to blocked Valve
3	8	Reverse Flow	Reverse flow detected
4	16	Flow Setpoint not reached	Setpoint cannot be reached within 15 min during flow control
5	32	Flow with closed valve	Flow is measured but valve is closed
6	64	Flow actual exceeds flow nominal	Actual flow exceeds the design nominal flow
7	128	Flow measurement error	Air in System, error occurred during flow measurement
8	256	Remote Temperature not OK	No connection to remote temperature Sensor
9	512	Flow body temperature not OK	Error with embedded sensor
10	1024	Communication to sensor interrupted	Internal communication to the flow sensor interrupted
11	2048	Freeze Warning	Measured Temperature and Glycol concentration indicate ice can build up
12	4096	Glycol Detected	Glycol was detected in a MID application
13	8192	Power Setpoint not reached	Setpoint cannot be reached within 15 min during Power Control

EV4.2 – Error State, AV[140] (highest code 524287)

Bit	Value	Message	Diagnostics
0	1	No Communication with Actuator	Communication with Actuator not Possible
1	2	Gear Disengaged	Gear button is pressed
2	4	Actuator cannot move	Mechanical Overload due to blocked Valve
3	8	Reverse Flow	Reverse flow detected
4	16	Flow Setpoint not reached	Setpoint cannot be reached within 15 min during flow control
5	32	Flow with closed valve	Flow is measured but valve is closed
6	64	Flow actual exceeds flow nominal	Actual flow exceeds the design nominal flow
7	128	Flow measurement error	Air in System, error occurred during flow measurement
8	256	Remote Temperature not OK	No connection to remote temperature Sensor
9	512	Flow body temperature not OK	Error with embedded sensor
10	1024	Communication to sensor interrupted	Internal communication to the flow sensor interrupted
11	2048	Freeze Warning	Measured Temperature and Glycol concentration indicate ice can build up
12	4096	Glycol Detected	Glycol was detected in a MID application
13	8192	Power Setpoint not reached	Setpoint cannot be reached within 15 min during Power Control
14	16384	Device end of life detected	MID Only. Sensor Module must be replaced
15	32768	Bus watchdog triggered	Timeout for the Bus Watchdog expired. Refer to AV[130] & MV[130]
16	65536	No Differential pressure detected	No differential pressure detected within 5 min during Pressure Control
17	131072	Differential pressure set point cannot be reached	Differential Pressure cannot be reached within 15 min during Pressure Control mode
18	262144	Minimum position applied	Min Position (27%) is applied if: -The valve is restarted - After a Power Failure - The manual Override was previously operated - -Switching from another Control mode -No dP is present at a flow rate <0.7% of V'nom

EV4 and EV4.2 – Non-Persistent errors

Bit	Value	Message	Diagnostics
1	2	Transit time measurement timeout	
2	4	Transit time out of range	
3	8	automatic gain controller at limit	
5	32	volume accumulation failed	
6	64	last heat/cold accumulation failed	
7	128	raw resistance measurement of sensor 1 (remote) or 2 (flow body) is invalid	
8	256	calculation error	
9	512	resistance measurement sensor 2 (flow body) at minimum	
10	1024	resistance measurement sensor 2 (flow body) at maximum	
11	2048	resistance measurement sensor 1 (remote) at minimum	
12	4096	resistance measurement sensor 1 (remote) at maximum	
13	8192	reverse flow detected	
14	16384	volumetric flow above high limit	
15	32768	volumetric flow not valid	

EV4 and EV4.2 – Persistent errors

Bit	Value	Message
0	1	resistance measurement sensor 2 (flow body) at minimum for several successive measurements (only set after device is legally commissioned)
1	2	resistance measurement sensor 2 (flow body) at maximum for several successive measurements (only set after device is legally commissioned)
2	4	resistance measurement sensor 1 (remote) at minimum for several successive measurements (only set after device is legally commissioned)
3	8	resistance measurement sensor 1 (remote) at maximum for several successive measurements (only set after device is legally commissioned)
4	16	Permanent communication error with non-volatile memory (SPI)
5	32	Integrity check of program code failed
6	64	Integrity check of parameter failed
7	128	Power failure occurred after legal commissioning is completed (only for MID devices)
8	256	Data format in non-volatile memory does not match with data format in Sensor-uC software
9	512	Integrity check of data in non-volatile memory failed
10	1024	Maximum error counter value reached

A.2 ePIV

ePIV2 – MI[106] (highest code 15)

Bit	Value	Message	Diagnostics
0	1	OK	OK
1	2	Actuator cannot move	Mechanical Overload due to blocked Valve
2	4	Gear disengaged	Gear button is pressed
3	8	Mechanical Travel Increased	The actuator has been moved outside the adapted working range

ePIV3 – MI[110] (highest code 15)

Bit	Value	Message	Diagnostics
0	1	OK	OK
1	2	Actuator cannot move	Mechanical Overload due to blocked Valve
2	4	Gear disengaged	Gear button is pressed
3	8	Mechanical Travel Increased	The actuator has been moved outside the adapted working range

ePIV4 – Error State, AV[140] (highest code 65535)

Bit	Value	Message	Diagnostics
0	1	No Communication with Actuator	Communication with Actuator not Possible
1	2	Gear Disengaged	Gear button is pressed
2	4	Actuator cannot move	Mechanical Overload due to blocked Valve
3	8	Reverse Flow	Reverse flow detected
4	16	Flow Setpoint not reached	Setpoint cannot be reached within 15 min during flow control
5	32	Flow with closed valve	Flow is measured but valve is closed
6	64	Flow actual exceeds flow nominal	Actual flow exceeds the design nominal flow
7	128	Flow measurement error	Air in System, error occurred during flow measurement
9	512	Flow body temperature not OK	Error with embedded sensor
10	1024	Communication to sensor interrupted	Internal communication to the flow sensor interrupted
11	2048	Freeze Warning	Measured Temperature and Glycol concentration indicate ice can build up
12	4096	Glycol Detected	Glycol was detected in a MID application
15	32768	Bus watchdog triggered	Timeout for the Bus Watchdog expired. Refer to AV[130] & MV[130]

ePIV4 – Non-Persistent errors

Bit	Value	Message
1	2	transit time measurement timeout
2	4	transit time out of range
3	8	automatic gain controller at limit
5	32	volume accumulation failed
6	64	last heat/cold accumulation failed
7	128	raw resistance measurement of sensor 1 (remote) or 2 (flow body) is invalid
8	256	calculation error
9	512	resistance measurement sensor 2 (flow body) at minimum
10	1024	resistance measurement sensor 2 (flow body) at maximum
11	2048	resistance measurement sensor 1 (remote) at minimum
12	4096	resistance measurement sensor 1 (remote) at maximum
13	8192	reverse flow detected
14	16384	volumetric flow above high limit
15	32768	volumetric flow not valid

ePIV4 – Persistent errors

Bit	Value	Message
0	1	resistance measurement sensor 2 (flow body) at minimum for several successive measurements (only set after device is legally commissioned)
1	2	resistance measurement sensor 2 (flow body) at maximum for several successive measurements (only set after device is legally commissioned)
2	4	resistance measurement sensor 1 (remote) at minimum for several successive measurements (only set after device is legally commissioned)
3	8	resistance measurement sensor 1 (remote) at maximum for several successive measurements (only set after device is legally commissioned)
4	16	Permanent communication error with non-volatile memory (SPI)
5	32	Integrity check of program code failed
6	64	Integrity check of parameter failed
7	128	Power failure occurred after legal commissioning is completed (only for MID devices)
8	256	Data format in non-volatile memory does not match with data format in Sensor-uC software
9	512	Integrity check of data in non-volatile memory failed
10	1024	Maximum error counter value reached

A.3 TEM

TEM4 – AV[140] (highest code 8191)

Bit	Value	Message	Diagnostics
3	8	Reverse Flow	Reverse flow detected
6	64	Flow Actual exceeds Flow Nominal	Actual flow exceeds the design nominal flow
7	128	Flow Measurement Error	Air in the System.
8	256	Remote Temperature not OK	No connection to the Remote Sensor
9	512	Flow body Temperature not OK	No connection to the embedded sensor
10	1024	Communication to Sensor interrupted	Internal Communication to the flow sensor interrupted
11	2048	Freeze Warning	Measured Temperature and Glycol concentration indicate ice can builds up
12	4096	Glycol Detected	Glycol was detected in a MID application

TEM 4.2 – MV[103] Status Sensor (highest code 63)

Bit	Value	Message	Diagnostics
1	2	OK	OK
2	4	Flow Measurement Error	Air in the System.
3	8	Flow body Temperature not OK	No connection to the embedded sensor
4	16	Remote Temperature not OK	No connection to the Remote Sensor
5	32	Communication to Sensor interrupted	Internal Communication to the flow sensor interrupted

TEM 4.2 – MV[104] Status Flow (highest code 63)

Bit	Value	Message	Diagnostics
1	2	OK	OK
2	4	Actual Flow exceeds nominal Flow	
5	32	Reverse Flow	Reverse Water Flow Detected

TEM 4.2 – MV[105] Status Media (highest code 63)

Bit	Value	Message	Diagnostics
1	2	OK	OK
2	4	Glycol Detected	Glycol was detected in a MID application
3	8	Freeze Warning	Measured Temperature and Glycol concentration indicate ice can builds up

Appendix B. Release History

Release	Date	Changes
1.5	2026-09-23	Fixed Energy Valve EV3 decoding of codes that include Freeze Warning (2048). Fixed TEM 4.2 (MV[103], MV[104], MV[105]) decoding, which stopped after its first row. Fixed the Code Type list showing duplicate entries after changing Product. Every version and bit was re-tested.
1.4	2026-09-22	Added automatic update checking through Belimo's update source. Added Check for Updates and Download Fresh Copy buttons to the About screen. Updates save through a normal Windows Save As dialog and never touch your current copy. The About screen's version line now reads the live release number. The About screen also has a Download User Manual button, which fetches this manual.